Karetao bench wiring: STM32F405 hub, TCA9548A mux, AS5600 encoders, CAN to six MKS drives, SPI to the Raspberry Pi 5.

Signals are connected by net labels of the same name. Power symbols: +3V3, +5V, GND, VMOT_SW (motor supply after the E-stop).

Confirmed by the documents: STM32 pins (ARM_SENSOR_HUB.md), mux address 0x70 and axis N on channel N+2, encoder fitted per axis,

drive models and CAN IDs 0x001 to 0x006 (SYSTEM_CONFIGURATION.md 2.1, 2.2), Pi header pins 19/21/23/24/22 (Pi README), 12 to 24 V drive supply (MKS manual).

Provisional (not stated in any document): STM32 board type and its 5 V input; TJA1050 S pin tie; I2C pull-up values, location and rails

(main bus drawn to +3V3; channel SCx/SDx pull-ups expected on the breakouts, rail not recorded); mux supply (drawn +3V3):

mux RESET pull-up (no MCU line, FW-12); AS5600 DIR tied low; termination placement;

motor supply voltage; the E-stop itself (SAFE-01, ruling R25b: to be fitted); drive switch and motor phase wiring (Arctos drawing, bench check).

Decoupling on breakout boards is not drawn. Gripper parts (Nano, MCP2515, servo) are out of scope and not drawn (charter).

Generated by Hardware/bench-wiring/generate_schematic.py: edit the script, not this file, then regenerate. Run ERC in KiCad after any change.

J1
Raspberry Pi 5 header
1_3V3
2_5V
19_GPIO10_MOSI — SPI_MOSI
21_GPIO9_MISO — SPI_MISO
23_GPIO11_SCLK — SPI_SCK
24_GPIO8_CE0 — SPI_NSS
22_GPIO25_DATA_READY — DATA_READY
6_GND
20_GND
GND

+3V3
R3 4k7 — I2C1_SCL
+3V3
R4 4k7 — I2C1_SDA
+3V3
R5 10k — MUX_NRESET

R3, R4: I2C pull-ups; R5: mux reset pull-up. Values and location Provisional.

+3V3 +5V
U1
STM32F405RGT6 board
VIN_5V
3V3
NRST
PA13_SWDIO
PA14_SWCLK
GND
PB6_I2C1_SCL — I2C1_SCL
PB7_I2C1_SDA — I2C1_SDA
PB8_CAN1_RX — CAN_RX
PB9_CAN1_TX — CAN_TX
PA15_SPI3_NSS — SPI_NSS
PC10_SPI3_SCK — SPI_SCK
PC11_SPI3_MISO — SPI_MISO
PC12_SPI3_MOSI — SPI_MOSI
PA8_DATA_READY — DATA_READY
NRST
SWDIO
SWCLK
GND

+3V3
U2
TCA9548A
I2C1_SDA
I2C1_SCL
MUX_NRESET
VCC
SDA
SCL
nRESET
A0
A1
A2
GND
SC0 — SCL_CH0
SD0 — SDA_CH0
SC1
SD1
SC2 — SCL_CH2
SD2 — SDA_CH2
SC3 — SCL_CH3
SD3 — SDA_CH3
SC4 — SCL_CH4
SD4 — SDA_CH4
SC5 — SCL_CH5
SD5 — SDA_CH5
SC6 — SCL_CH6
SD6 — SDA_CH6
SC7 — SCL_CH7
SD7 — SDA_CH7
GND

A2 A1 A0 = L L L: address 0x70

J2
SWD header
3V3
SWDIO
SWCLK
NRST
GND
+3V3
GND

+5V
U3
AS5600
SCL_CH0
SDA_CH0
VDD
SCL
SDA
DIR
OUT
PGO
GND
GND
GND
Bench sensor on SD0 (excluded from the arm)

+5V
U4
AS5600
SCL_CH2
SDA_CH2
GND
GND
Axis 0, Joint X: NO ENCODER FITTED (sensor on hand; NOT_FITTED, SYS-ENC-002)

+5V
U5
AS5600
SCL_CH3
SDA_CH3
GND
GND
Axis 1, Joint Y, encoder fitted

+5V
U6
AS5600
SCL_CH4
SDA_CH4
GND
GND
Axis 2, Joint Z, encoder fitted

+5V
U7
AS5600
SCL_CH5
SDA_CH5
GND
GND
Axis 3, Wrist Joint 1: NO ENCODER FITTED (NOT_FITTED, SYS-ENC-002)

+5V
U8
AS5600
SCL_CH6
SDA_CH6
GND
GND
Axis 4, Wrist Joint 2, encoder fitted

+5V
U9
AS5600
SCL_CH7
SDA_CH7
GND
GND
Axis 5, Wrist Joint 3, encoder fitted

+5V
U10
TJA1050
VCC
TXD
RXD
VREF
S
GND
CANH — CAN_H
CANL — CAN_L
CAN_TX
CAN_RX
GND

CAN_H
R1 120R
CAN_L

R1: termination at the hub end (Provisional)

U10 TJA1050 on +5V: RXD swings to 5 V into PB8 (FT per DS8626 Table 7: 5 V tolerant only while the STM32 is powered; no pull-up). S tied low (Provisional).

CAN bus: CAN_H and CAN_L daisy-chained U10 -> M1 ... M6 (and J3 when used), 120R at each end (R1, R2); every drive's own terminator off

(57D DIP 1, 42D shorting cap). Drives and adapter share GND with the hub (MKS manual p. 18). Drive blocks: bottom of the sheet.

PS2
Logic PSU 5 V (Provisional)
5V
GND
+5V
GND

J3
USB-CAN adapter (MKS Canable)
CANH — CAN_H
CANL — CAN_L
GND
GND

J3: USB-CAN adapter for bring-up phase 7a. Its GND is connected (the Arctos drawing leaves it floating).

CAN_H
R2 120R
CAN_L

R2: far-end termination (Provisional)

AS5600s on +5V (Lenton Neale, 2026-10-15): in 5 V mode a logic high needs about 0.7 x VDD, so each channel's SCx/SDx must be pulled to the sensor rail, not +3V3. Channel pull-ups are not drawn: confirm the breakout pull-ups and all rails in BRING_UP phase 2 (B3).

Motor supply: PSU -> rocker switch (fitted) -> two-pole E-stop (to be fitted, in line after the rocker) -> VMOT_SW to every drive VIN+; drive VIN- to GND

PS1
Motor PSU 24 V (Provisional)
V+
V-
SW2
Rocker switch (fitted)
SW1
E-STOP (to be fitted)
IN1
IN2
OUT1
OUT2
VMOT_SW
GND
PWR_FLAG VMOT_SW

PWR_FLAG: VMOT_SW is fed through the passive switches

Drive I/O, from the original Arctos wiring. Home and limit switch wiring and motor phase order are Provisional until confirmed on the bench (BRING_UP phase 1).

Phase A is crossed at every drive (drive A+ to motor A-), as in the Arctos drawing; it only sets the turning direction. Switch modules take 5 V from the drive's EVCC/EGND (20 mA budget).

Axis 0: Joint X (Base)

CAN ID 0x001, SERVO57D; motor NEMA 23, 2.8 A

Switches on IN_1 / IN_2: DIP 2, 3 ON: EVCC 5 V, DIP 1 (terminator) OFF.

Phase A crossed (Lenton recalls doing this on the 57D; confirm)

M1
MKS SERVO57D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
IN_2
OUT_1
OUT_2
RS485_A
RS485_B
A+ — AX0_A+
A- — AX0_A-
B+ — AX0_B+
B- — AX0_B-
AX0_EVCC
AX0_EGND
AX0_IN_1
AX0_IN_2

MOT1
NEMA 23, 2.8 A
AX0_A-
AX0_A+
AX0_B+
AX0_B-
A+
A-
B+
B-

SW11
WSH231 (home)
AX0_EVCC
AX0_EGND
AX0_IN_1
AX0_IN_2
VCC
GND
N_OUT
S_OUT

Axis 1: Joint Y (Shoulder)

CAN ID 0x002, SERVO57D; motor NEMA 23, 2.8 A

Switches on IN_1 / IN_2: DIP 2, 3 ON: EVCC 5 V, DIP 1 (terminator) OFF.

Phase A crossed (Lenton recalls doing this on the 57D; confirm)

M2
MKS SERVO57D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
IN_2
OUT_1
OUT_2
RS485_A
RS485_B
A+ — AX1_A+
A- — AX1_A-
B+ — AX1_B+
B- — AX1_B-
AX1_EVCC
AX1_EGND
AX1_IN_1
AX1_IN_2

MOT2
NEMA 23, 2.8 A
AX1_A-
AX1_A+
AX1_B+
AX1_B-
A+
A-
B+
B-

SW12
KY-003 (left limit)
AX1_EVCC
AX1_EGND
AX1_IN_1
Vin
GND
S

SW13
KY-003 (right limit)
AX1_EVCC
AX1_EGND
AX1_IN_2
Vin
GND
S

Axis 2: Joint Z (Elbow)

CAN ID 0x003, SERVO42D; motor NEMA 17, 2.1 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

M3
MKS SERVO42D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
A+ — AX2_A+
A- — AX2_A-
B+ — AX2_B+
B- — AX2_B-
AX2_EVCC
AX2_EGND
AX2_EVCC
AX2_EN
AX2_DIR

MOT3
NEMA 17, 2.1 A
AX2_A-
AX2_A+
AX2_B+
AX2_B-
A+
A-
B+
B-

SW14
KY-003 (left limit)
AX2_EVCC
AX2_EGND
AX2_EN
Vin
GND
S

SW15
KY-003 (right limit)
AX2_EVCC
AX2_EGND
AX2_DIR
Vin
GND
S

Axis 3: Wrist Joint 1 (Wrist 1)

CAN ID 0x004, SERVO42D; motor NEMA 17, 1.3 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

M4
MKS SERVO42D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
A+ — AX3_A+
A- — AX3_A-
B+ — AX3_B+
B- — AX3_B-
AX3_EVCC
AX3_EGND
AX3_EVCC
AX3_EN
AX3_DIR

MOT4
NEMA 17, 1.3 A
AX3_A-
AX3_A+
AX3_B+
AX3_B-
A+
A-
B+
B-

SW16
WSH231 (home)
AX3_EVCC
AX3_EGND
AX3_EN
AX3_DIR
VCC
GND
N_OUT
S_OUT

Axis 4: Wrist Joint 2 (Wrist 2)

CAN ID 0x005, SERVO42D; motor NEMA 17, 1.2 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

M5
MKS SERVO42D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
A+ — AX4_A+
A- — AX4_A-
B+ — AX4_B+
B- — AX4_B-
AX4_EVCC
AX4_EGND
AX4_EVCC
AX4_EN
AX4_DIR

MOT5
NEMA 17, 1.2 A
AX4_A-
AX4_A+
AX4_B+
AX4_B-
A+
A-
B+
B-

SW17
KY-003 (left limit)
AX4_EVCC
AX4_EGND
AX4_EN
Vin
GND
S

SW18
KY-003 (right limit)
AX4_EVCC
AX4_EGND
AX4_DIR
Vin
GND
S

Axis 5: Wrist Joint 3 (Wrist 3)

CAN ID 0x006, SERVO42D; motor NEMA 17, 1.2 A

9EH port mapping on: left -> En, right -> Dir; COM to EVCC. IN_1 unused. Terminator cap off.

Phase A crossed per the Arctos drawing (not confirmed)

M6
MKS SERVO42D
VMOT_SW
CAN_H
CAN_L
GND
VIN+
CANH
CANL
VIN-
GND
EVCC
EGND
COM
EN
DIR
STP
IN_1
A+ — AX5_A+
A- — AX5_A-
B+ — AX5_B+
B- — AX5_B-
AX5_EVCC
AX5_EGND
AX5_EVCC
AX5_EN
AX5_DIR

MOT6
NEMA 17, 1.2 A
AX5_A-
AX5_A+
AX5_B+
AX5_B-
A+
A-
B+
B-

SW19
WSH231 (home)
AX5_EVCC
AX5_EGND
AX5_EN
AX5_DIR
VCC
GND
N_OUT
S_OUT

Source: Hardware/bench-wiring/generate_schematic.py
System wiring at connector level: Provisional items noted on the sheet
Karetao
Sheet: /
File: karetao_bench_wiring.kicad_sch

Title: Karetao bench wiring

Size: A2 | Date: 2026-10-15 | **Rev: 0.3**
KiCad E.D.A. 10.0.7 | Id: 1/1